Q1: For the following lines of code

```
ld     x2,    0(x10)
add    x3,    x2,    x4
ld     x12,   4(x10)
or     x13,   x12,   x14
```

1. Draw a multi-cycle clock diagram. Assume we have hazard detection and forwarding units.

Instr	0	1	2	3	4	5	6	7	8
ld	F	D	E	M	W				
add→ NOP		F	D	-	-	-			
add			F	D	E	M	W		
ld				F	D	E	M	W	
or → NOP					F	D	-	-	-
or						F	D	E	M

2. Draw a multi-cycle clock diagram. Assume we have hazard detection unit but no forwarding unit.

Instr	0	1	2	3	4	5	6	7	8		
ld	F	D	E	M	W						
add→ NOP		F	D	-	-	-					
add→ NOP			F	D	-	-	-				
add				F	D	E	M	W			
ld					F	D	E	M	W		
or → NOP						F	D	-	-	-	
or → NOP							F	D	-	-	-
or								F	D	E	M

3. Assume we don't have hazard detection or forwarding units in data-path. Re-write code and resolve hazards using nops.

Sol:

```
ld      x2,     0(x10)
NOP
NOP
add     x3,     x2,     x4
ld      x12,    4(x10)
NOP
NOP
or      x13,    x12,    x14
```